Divyansh Shukla

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Education

Vellore Institute of Technology Expected 2027
B.Tech, Computer Science and Engineering (Artificial Intelligence & Machine Learning) *CGPA: 9.03 / 10*

Experience

Power Pulse India 2026 – Present
Machine Learning Engineer Intern *India*

- Trained YOLOv8n on the FLIR ADAS v2 thermal dataset to detect 6 road-safety classes: person, bicycle, car, motorcycle, bus, and truck.
- Ran multi-hour GPU training jobs on Modal with persistent volumes, stored secrets, and Weights & Biases logging, and added checkpoint resume so an interrupted run picks up where it stopped.
- Tuned training for single-channel thermal input: no hue or saturation augmentation, higher classification loss weight to cut background confusion. Evaluation reports per-class AP50 and AP50-95 against a mAP50 0.59 / mAP50-95 0.36 baseline.

Certifications

NPTEL: Applied Accelerated AI, Deep Learning with Computer Vision | **Coursera**: Deep Learning Specialization | **AWS**: Cloud Architecting (trained)

Technical Skills

Languages: Python, C++, CUDA
ML Frameworks: PyTorch, PyTorch Lightning, Hugging Face Transformers, TRL, Ultralytics (YOLO), Stable-Baselines3, vLLM
Deep Learning: diffusion models, flow matching, transformers, attention mechanisms (RoPE, AdaLN-Zero), classifier-free guidance, object detection, reinforcement learning (PPO), vision-language models
LLMs & Agents: prompt engineering, structured output extraction, function calling, agentic workflows (planning, memory, tool use), LLM evaluation
RAG & Retrieval: retrieval-augmented generation (RAG), vector databases, pgvector, ChromaDB, embeddings
AI Developer Tools: Claude Code, Gemini Antigravity, Gemini API
Simulation & Robotics: MuJoCo, Gymnasium, OpenCV, Raspberry Pi, embedded/edge inference
Backend & Infrastructure: FastAPI, Django, Flask, Modal, Weights & Biases, Hydra, Git, Linux, NumPy, pandas

Projects

RoPE-DiT: Diffusion Transformer with 2D RoPE github.com/divyanshuklai/Custom2DRoPEDiT

- Wrote a ∼5M-parameter Diffusion Transformer from scratch in PyTorch: patchifier, timestep and class-label embedders, AdaLN-Zero conditioning, and a custom multi-head attention module.
- Built a 2D rotary positional embedding that rotates queries and keys by each patch's height and width coordinates, splitting the head dimension into row and column halves so attention sees relative 2D distance.
- Trained on CIFAR-10 with a rectified flow objective and sampled with a hand-written Euler solver and classifier-free guidance.

Zero-Shot ICE RMA: Quadruped Locomotion RL github.com/divyanshuklai/zero-shot-ice-rma

- Built a MuJoCo environment for the Unitree Go1 quadruped with 30-dim proprioceptive observations, 12-dim joint-position actions, fractal terrain, and the Rapid Motor Adaptation bioenergetics reward with curriculum-scheduled penalties.
- Debugged a PPO baseline that ran 42M steps without learning a gait, tracing the exploding value loss to unnormalized observations and penalty dominance.

Fashion Search: Conversational Product Discovery github.com/divyanshuklai/fashion_search

- Built a FastAPI chat app answering natural-language product queries across 10 Indian e-commerce stores with Tavily search, Gemini 2.5 Flash extraction, and BeautifulSoup scraping.
- Wrote a priority-based image and price extractor that tries JSON-LD product schema, then Open Graph tags, then keyword-matched image tags, as the baseline for an agentic version with planning, memory, and tool use.

Achievements

Won the internal (intra-campus) round of the Smart India Hackathon at Vellore Institute of Technology, 2024.